WHITE PAPER · CLINICAL AI SAFETY

The Human-in-the-Loop Illusion

Architectural Safety Guarantees for Clinical AI

Humans keep authority over consequential decisions. Architecture takes responsibility for detecting specified hazards.

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For clinical leaders, regulators, patient safety officers and engineers. Includes an author-designed elder care example, bench results and disclosure of commercial interests.

EXECUTIVE SUMMARY

Human review is the wrong tool for detecting rare AI errors

A reviewing clinician provides accountability. Their presence does not establish that rare errors will be detected.

Clinical AI policy repeatedly relies on a person to catch a system's mistakes. Decades of human factors research challenge that assumption: sustained monitoring degrades, trust encourages acceptance, and frequent low-value warnings teach people to dismiss the next one. A meta-analysis found drug interaction alerts were overridden 90 percent of the time.

Keep human authority

People decide in context, weigh information the system cannot see, explain decisions and answer for outcomes.

Move detection into architecture

Screen every input, validate every output and escalate specified hazards without depending on a reviewer noticing them.

Require evidence for both

Measure AI failure rates and human detection rates alongside the controls intended to close the gap.

The proposal combines five mature tools: safety envelopes, input guards and output validators, fail-closed defaults, uncertainty with coverage guarantees, and runtime verification. Their guarantees remain bounded by their specifications and assumptions.

For clinical leaders, regulators, patient safety officers and engineers.

POLICY · ACCOUNTABILITY

Every regime puts a person between AI and the patient

Oversight rules assign authority well. They do not, by themselves, demonstrate reliable error detection.

EU · Article 14

The EU AI Act requires effective human oversight, including awareness of over-reliance. High-risk obligations apply from December 2027 for standalone systems and August 2028 for systems embedded in devices.

US federal · FDA clinical decision support

The regulatory distinction turns partly on whether a healthcare professional can independently review the basis for a recommendation.

US states · California and Texas

California SB 1120 and Texas SB 815 require licensed professionals for adverse medical necessity determinations.

Policy strip · Jurisdictions and requirements described in the paper; dates shown where supplied.

The missing evidentiary step

Ability to review and responsibility to decide do not demonstrate the ability to find rare mistakes in clinical work. Treating a reviewer's presence as proof that errors get caught leaves that performance untested.

THE CENTRAL DISTINCTION

One person is asked to do two different jobs

“Human oversight” bundles a legitimate decision-making role with a demanding monitoring task. Those roles need different safeguards and different evidence.

The reviewing clinician

One person, two responsibilities

↓

AUTHORITY · STRONG FIT

Decide in context

Weigh patient preferences, circumstances and information outside the system. Choose a course of action and answer for the outcome.

DETECTION · WEAK FIT

Find rare errors

Spot the unusual wrong answer in a stream of mostly correct output while managing interruptions, workload and time pressure.

Role split · The distinction concerns task design, not the competence or commitment of clinicians.

Presence is an inadequate measure

Policy rarely separates these jobs. A person’s presence satisfies the authority requirement and is silently assumed to satisfy detection. Without measured detection performance, the safety case rests on an unverified assumption.

EVIDENCE · HUMAN FACTORS

Seven decades of evidence: vigilance fades and trust in automation grows

1948 · Mackworth

Radar operators' detection fell roughly 10 to 15 percent within the first half hour. Sustained attention degrades.

1983 · Bainbridge

Automation leaves people responsible for intervening when reduced practice and engagement make intervention harder.

2000 · Parasuraman, Sheridan and Wickens

Automation's allocation of tasks changes what people monitor and how ready they are to take over.

2010 · Parasuraman and Manzey

Complacency and automation bias affect novices and experts, producing both omission and commission errors.

2012 · JAMIA review

Across 74 healthcare studies, workload, complexity, time pressure and trust shaped automation bias. Showing information instead of a recommendation helped.

Omission

Failing to act because the system did not flag a problem.

Commission

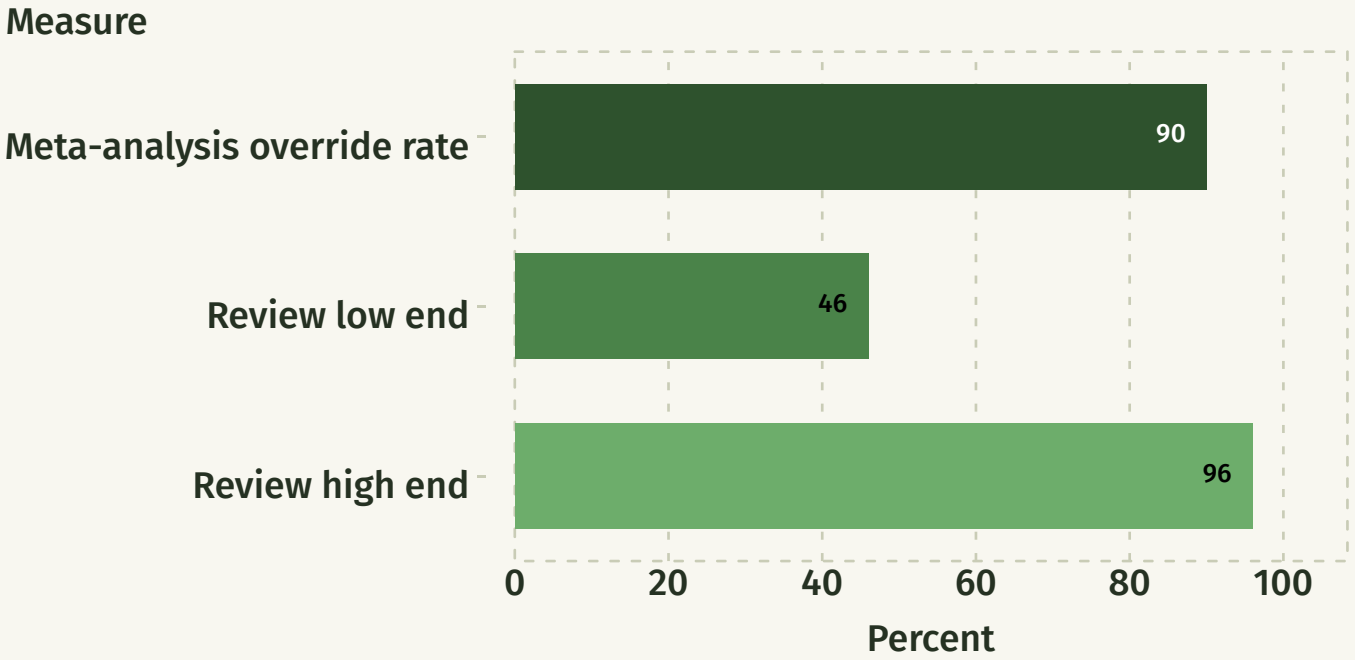
Following incorrect automated advice despite conflicting information.

Evidence timeline · These findings describe recurring task vulnerabilities, not an inevitable error on every review.

EVIDENCE · ALERT FATIGUE

Alert fatigue: when most alerts are noise, the rare real one looks the same

A warning channel loses discriminating power when clinicians repeatedly encounter alerts they judge irrelevant. The next critical warning arrives through that same learned dismissal pathway.



Override figures · The 2024 meta-analysis included 16 studies: 90 percent of drug-drug interaction alerts were overridden, with 95% CI 85 to 95. The earlier 23-study review reported average override rates from 46 to 96 percent. These are different evidence summaries, not a time trend.

Alarm harm is consequential

The Joint Commission reported 98 alarm-related events between 2009 and 2012, 80 resulting in death, in its 2013 alert.

Some overrides in the earlier review were clinically inappropriate. Override frequency alone does not establish which individual dismissal was wrong, but it undermines a safety model that treats every displayed alert as an effective intervention.

CASE · UCSF

At UCSF every human was present and none stopped a 38.5x overdose

38.5×

A teenage patient received 38.5 times the intended antibiotic dose, as told in Wachter's *The Digital Doctor*.

The ordering warning resembled the routine dosing warnings clinicians had learned to click through. The medication then passed through successive opportunities for interception.

1	2	3	4	5
Order warning	**Pharmacist**	**Dispensing robot**	**Technician**	**Nurse**
Missed chance	Missed chance	Missed chance	Missed chance	Missed chance

Safeguard chain · Sequence of interception opportunities described in the account. This was a medication-system failure, not evidence that a clinical AI caused the overdose.

Multiple reviewers are not necessarily independent protection

The case illustrates the vulnerability of a system that assumes someone will notice. Familiar warnings and successive handoffs can coexist with a catastrophic error. Adding people to the chain does not establish that the chain contains an effective, independent safety check.

FRAMEWORK · SYMMETRIC FAILURE ANALYSIS

Symmetric failure analysis holds the overseer to the same standard as the AI

For every failure mode, ask how the AI fails, how the human overseer fails, and which architectural control closes the gap. Neither component receives an exemption from evidence.

Failure mode	AI component	Human overseer	Architectural mitigation
Missed danger	Model does not flag risk	Vigilance decrement, omission	OR-combined gates, abstention
Following wrong output	Plausible but wrong output	Automation bias	Output validation, evidence shown
Alert dismissal	Not applicable to deterministic checks	High override rates	Fewer, better alerts; critical bypass
Fatigue over time	Consistent	Detection declines	Automated escalation
Cascading failure	Silent propagation	Each assumes another checked	Independent verified layers
Drift	Behavior shifts	Overseers adapt	Monitor distributions and override rates

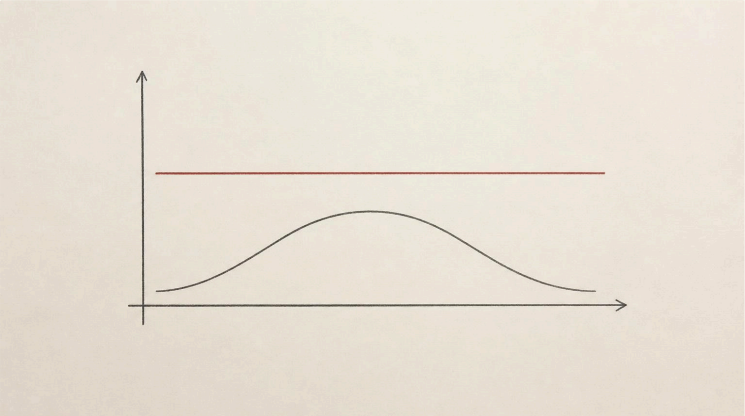
Failure-mode matrix · “Consistent” means the automated check does not tire; it does not mean an AI component is invariant to drift or universally correct.

Independent gates combined with OR logic escalate when any gate identifies a hazard. Independence must be established: several layers sharing one blind spot do not close the gap.

CONCEPTUAL MODEL · NOT MEASURED

No reliability level makes human detection dependable

A noisier AI breeds alert fatigue. A more reliable AI breeds complacency. Neither extreme establishes dependable human detection.



Concept sketch · Horizontal axis: AI reliability, lower to higher. Vertical axis: detection performance. Lines: Human detection (curve) and Architectural check (flat line). Conceptual only; no measured values.

The shapes express the argument, not empirical effect sizes or a universal performance ranking. Architectural checks execute consistently only within their specified operating assumptions.

The evidence should have three columns

AI failures	Human detection	Controls
How often does the system fail?	How often are those failures actually caught?	What independently prevents propagation?

The second column is almost never filled in. A control should run on the ten-thousandth input as on the first.

PROPOSED DIVISION OF LABOUR

Architecture detects, humans decide

Detection should deliver the right cases to people who retain authority over consequential decisions.

Architecture

- Screen every input
- Validate every output
- Measure uncertainty
- Monitor drift

People

- Judge in context
- Decide what to do
- Explain the decision
- Answer for outcomes

Architecture → targeted escalation → human decision

Send evidence and the reason for escalation, rather than a verdict that invites uncritical acceptance.

Make oversight a designed task

Fewer, better alerts preserve the distinction between routine processing and cases that require attention. Measure whether escalated cases reach the right person and whether the response is appropriate under real working conditions.

This is not autonomy. People take on more consequential decisions because the right cases reach them, rather than being asked to read every output and detect every exceptional failure.

SAFETY ARCHITECTURE · CONDITIONAL GUARANTEES

Five architectural guarantees hold regardless of attention

1 Deterministic safety envelope

A simple verifiable controller bounds a complex one: the Simplex pattern (Sha, 2001).

2 Unconditional guards and validators

Check every input and output; independent OR gates escalate any detected hazard.

3 Fail-closed defaults

Outages never look like an all-clear; safety detection runs locally without a network.

4 Uncertainty with coverage guarantees

Conformal coverage depends on assumptions; uncertainty triggers abstention and escalation.

5 Runtime verification and traceability

Verify agent actions and preserve an append-only hash-chained audit log.

$$P\{Y \in C(X)\} \geq 1 - \alpha$$

Conformal coverage is typically marginal and depends on assumptions such as exchangeability; it is not a per-patient guarantee or protection against arbitrary distribution shift.

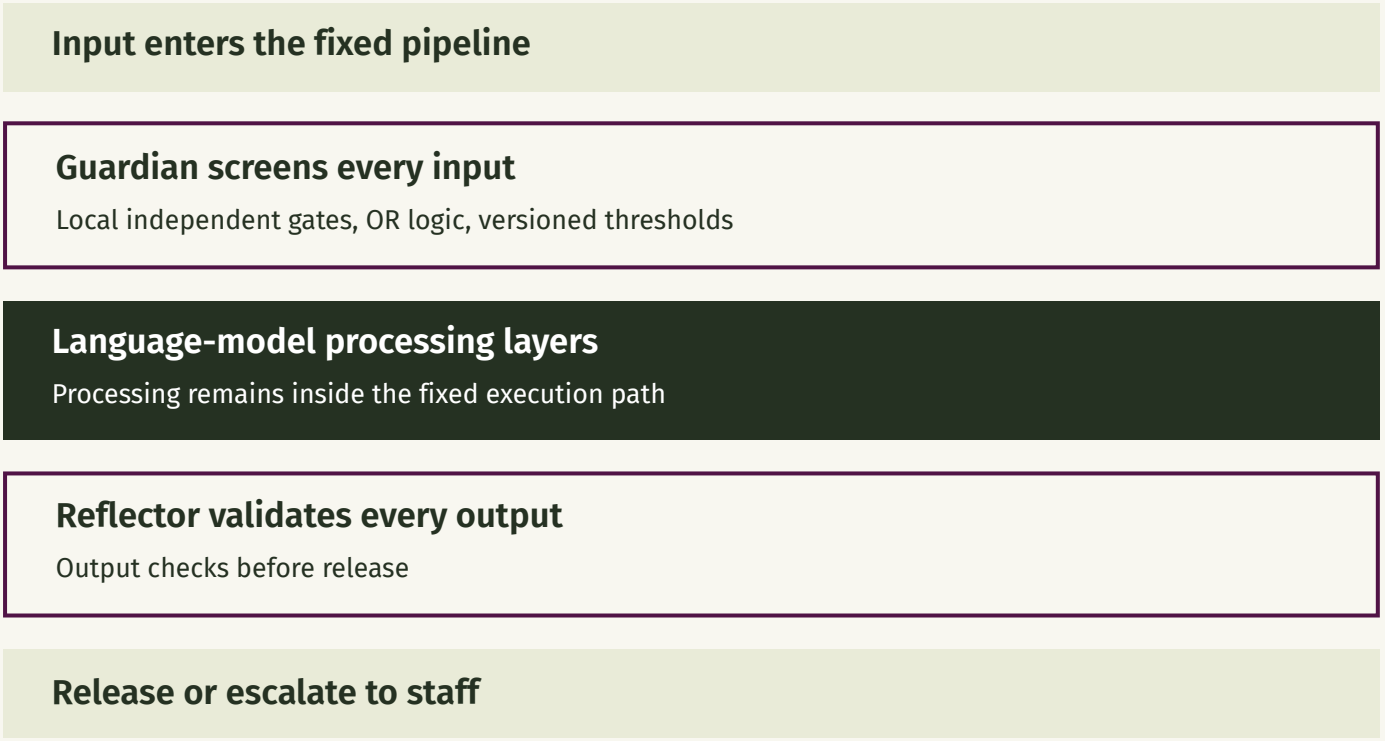


Envelope · Runtime checks surround processing; escalation goes to a person, and the audit log records the path. Guarantees cover specified properties, not all clinical harm.

WORKED PATTERN · ELDER CARE

Worked pattern: a deterministic pipeline in elder care

The author designed this system, which is the subject of a pending patent and a 2026 medRxiv preprint. The figures below are bench results, not a clinical study.



Five-layer schematic · Functional grouping, not a complete implementation specification. Staff respond to escalations and set criteria; nobody reads every conversation.

3,720
Bench scenarios

500
Crisis scenarios, all detected

<30 ms
Reported bench latency

The fixed pipeline cut execution paths from more than seven to two. Field performance still needs prospective validation; detection of all crisis scenarios in this bench set does not establish clinical sensitivity.

ASSURANCE · TEST BEFORE DEPLOYMENT

Testing is an ethical standard for safety architecture

A safety claim needs a repeatable test, a defined failure condition and an honest account of what was measured.



Assurance loop · Every change returns through testing; threshold review does not bypass validation.

Make the test corpus challenge the specification

Write safety tests first, covering explicit, indirect, buried, dialect and false-alarm scenarios. Keep a held-out adversarial corpus for testing, never tuning. Run tests on every change and on target hardware; a recall drop fails the build.

Make changes accountable

Version thresholds with their approver and supporting evidence. Pre-register expected effects so evaluation does not become a retrospective justification for the change.

Report the measurement, not the aspiration

Name the test set and its size, label bench results, maintain one source of truth for metrics and state goals as goals. Disclose interests. Prospective field validation remains necessary where a bench test cannot reproduce clinical conditions.

POLICY PROPOSALS · LIMITS · DISCLOSURE

Four policy recommendations, and the limits of this argument

01 Require evidence that oversight works

Measure detection under realistic staffing, volume and shift length. A named reviewer is an accountability mechanism, not a performance result.

02 Reward architectural safety

Create regulatory incentives for architectural controls, for example within FDA predetermined change control plans.

03 Align liability with autonomy

Clinicians should not become liability sinks for decisions they cannot meaningfully control (Mello and Guha, *NEJM*, 2024).

04 Require symmetric failure analysis

Make analysis of AI failure, human overseer failure and architectural mitigation a standard part of submissions.

Limits of the argument

Guarantees are only as good as their specification. The evidence comes largely from alerting tasks and does not establish identical effects across every clinical AI workflow. These recommendations are proposals requiring input from clinicians, patients, manufacturers and regulators.

Competing interests

The author has a commercial interest in the elder care system, which he designed and which is the subject of a pending patent and a 2026 medRxiv preprint. Its reported results are bench findings, not clinical validation.